

HDOC 12

Activity Description (Batches 15 & 16):

Prepare the algorithm and test case table. Then write, compile, execute, and test the C program against the prepared use cases. Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

- Check whether all its digits are unique. E.g. Number: 12345, Output: Unique. Number: 12234, Output: Not Unique
- Find the frequency of each digit from 0 to 9. E.g. Number: 112333, freq_0 = 0, freq_1 = 2, freq_2 = 1, freq_3 = 3...

Name: _____

SAP ID: _____

Batch: _____

1. Algorithm

1. Start

2. Declare variables for choice, num, temp, rem, isUnique, and an array freq[10] initialized to 0.
3. Display the menu options to the user.
4. Read the user's choice.
5. If choice is 1 or 2, prompt the user to input a positive integer num. Validate that $\text{num} > 0$.
6. Use a switch statement based on the value of choice.

7. Case 1 (Check unique digits):

- Initialize a tracking array `seen[10] = {0}`.
- Set `isUnique = 1` and `temp = num`.
- Loop while (`temp > 0`):
 - Calculate `rem = temp % 10`.
 - If `seen[rem] == 1`, set `isUnique = 0` and break the loop.
 - Else, set `seen[rem] = 1`.
 - Update `temp = temp / 10`.
- If `isUnique == 1`, display "Unique", else display "Not Unique".
- Break.

8. Case 2 (Digit frequencies):

- Initialize all elements of `freq[10]` to 0.

- Set temp = num.
- Loop while (temp > 0):
 - Calculate rem = temp % 10.
 - Increment freq[rem] by 1.
 - Update temp = temp / 10.
- Loop i from 0 to 9: display freq_i = freq[i].
- Break.

9. **Default Case:** Display an "Invalid choice" error message.

10. **Stop**

2. Test Case Table

TC ID	Scenario Description	Inputs (choice, num)	Expected Output
TC-01	Valid inputs (Unique digits)	Choice: 1 Num: 12345	Output: Unique
TC-02	Valid inputs (Repeating digits)	Choice: 1 Num: 12234	Output: Not Unique
TC-03	Frequency tracking (Option 2)	Choice: 2 Num: 112333	freq_1=2, freq_2=1, freq_3=3, rest=0
TC-04	Single digit number (Option 1)	Choice: 1 Num: 7	Output: Unique
TC-05	Invalid negative number input	Choice: 1 Num: -90	Error: Please enter a strictly positive integer.
TC-06	Invalid menu choice	Choice: 3	Invalid choice. Please choose 1 or 2.

3. C Program Source Code

```
#include <stdio.h>

int main() {
    int choice;
    long long num, temp;
    int rem, i;
```

```
int isUnique;
int seen[10] = {0};
int freq[10] = {0};

// Display Menu
printf("--- Digit Operations Menu ---\n");
printf("1. Check whether all digits are unique\n");
printf("2. Find the frequency of each digit (0-9)\n");
printf("Enter your choice (1 or 2): ");
scanf("%d", &choice);

// Validate Choice and Take Number Input
if (choice == 1 || choice == 2) {
    printf("Enter a positive integer: ");
    scanf("%lld", &num);

    if (num <= 0) {
        printf("Error: Please enter a strictly positive integer.\n");
        return 1; // Exit program due to invalid input
    }
}

// Switch Case Logic
switch (choice) {
    case 1:
        // Check for unique digits
        temp = num;
        isUnique = 1;

        // Re-initialize seen array to 0 just in case
        for(i = 0; i < 10; i++) seen[i] = 0;

        while (temp > 0) {
            rem = temp % 10;
            if (seen[rem] == 1) {
                isUnique = 0; // Digit repeated
                break;
            }
            seen[rem] = 1; // Mark digit as seen
            temp /= 10;
        }

        if (isUnique) {
            printf("Number: %lld, Output: Unique\n", num);
        } else {
            printf("Number: %lld, Output: Not Unique\n", num);
        }
    }
```

```

        break;

    case 2:
        // Find frequency of each digit
        temp = num;

        // Re-initialize freq array to 0
        for(i = 0; i < 10; i++) freq[i] = 0;

        while (temp > 0) {
            rem = temp % 10;
            freq[rem]++;
            temp /= 10;
        }

        printf("Number: %lld\n", num);
        for(i = 0; i < 10; i++) {
            printf("freq_%d = %d", i, freq[i]);
            if (i < 9) {
                printf(", ");
            }
        }
        printf("\n");
        break;

    default:
        printf("\nInvalid choice. Please choose 1 or 2.\n");
        break;
}

return 0;
}

```